

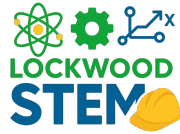
Introduction to Engineering Design

Aerospace Academy Version

Unit 7 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 7

Final Engineering Design Challenge

Full aerospace design challenge: problem definition, research, concept development, CAD/prototype, testing, iteration, documentation, and final review.

Grading Rubrics and Scoring Guide

Unit 7: Final Engineering Design Challenge

Assessment Purpose

Full aerospace design challenge: problem definition, research, concept development, CAD/prototype, testing, iteration, documentation, and final review. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 7 Score Summary

Category	Points Possible	Points Earned
Problem definition and mission need	10	
Research and requirements	10	
Concept development	10	
Design selection and justification	10	
CAD / technical design	15	
Prototype / model quality	10	
Testing and data analysis	15	
Iteration and risk response	10	
Final portfolio documentation	10	
Final design review	10	



Lockwood STEM

Academy of Advanced Aerospace Engineering

Page 3

Unit 7 Project Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete	
Problem Definition & Mission Need	10	Defines a meaningful aerospace problem with clear mission objective, stakeholders, criteria, and constraints.	Problem definition is clear and mostly complete.	Problem definition is vague or incomplete.	Problem is missing or unclear.	Introduction to Engineering Design Aerospace Academy Version
Research & Requirements	10	Research is relevant, summarized, and translated into design requirements and risks.	Research supports the design with adequate notes.	Research is limited or not connected to requirements.	Research missing.	
Concept Development	10	Multiple concepts are generated, compared, and refined using sketches and evidence.	Several concepts are considered and refined.	Few concepts or limited comparison.	No meaningful concept development.	
Design Selection & Justification	10	Uses clear criteria, decision matrix/trade-offs, and evidence to justify selection.	Selection is justified with some evidence.	Selection is mostly opinion-based.	No justification.	
CAD / Technical Design	15	CAD models/drawings are accurate, organized, parametric where appropriate, and ready for fabrication discussion.	CAD/drawings are complete with minor errors.	CAD/drawings are incomplete or inaccurate.	Missing technical design.	
Prototype / Model Quality	10	Prototype or model is appropriate, carefully made, and matches the intended test goals.	Prototype/model is complete and usable.	Prototype/model is incomplete or weakly connected to design.	Missing.	Rubrics
Testing & Data Analysis	15	Test plan is repeatable; data are complete; analysis connects results to requirements and design decisions.	Testing and analysis are adequate.	Testing is incomplete or data are weak.	No meaningful testing.	
Iteration & Risk Response	10	Redesign is specific, evidence-based, and addresses failures, risks, or unmet requirements.	Iteration is reasonable and explained.	Iteration is minor or not data-based.	No iteration.	
Final Portfolio Documentation	10	Portfolio is complete, professional, organized, and shows the full design process.	Portfolio is complete with minor gaps.	Portfolio is incomplete or hard to follow.	Missing portfolio.	

Design Review Presentation Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete
Engineering Story	20	Presentation clearly explains need, process, evidence, selected design, and next steps as one coherent engineering story.	Presentation explains main design process and final concept clearly.	Presentation is fragmented or focuses mostly on final idea.	Presentation lacks coherent explanation.
Use of Evidence	20	Uses criteria, constraints, research, sketches, matrix, data, and limitations to support claims.	Uses several pieces of evidence appropriately.	Evidence is limited, unclear, or weakly connected.	Claims are unsupported.
Visual Communication	20	Slides are clean, visual, readable, and include useful sketches/models/tables without overcrowding.	Slides are readable and include required visuals.	Slides are text-heavy, cluttered, or missing important visuals.	Slides missing or unreadable.
Team Delivery	20	All members contribute, speak professionally, stay within time, and respond to questions with engineering reasoning.	Most members contribute and delivery is clear.	Delivery is uneven, rushed, or lacks preparation.	Team is not prepared or does not present professionally.
Reflection and Next Steps	20	Clearly identifies limitations, risks, and specific next tests/redesigns.	Identifies reasonable limitations and next steps.	Next steps are vague or not connected to evidence.	No limitations or next steps.

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Rubrics

Student Self-Check Before Submission

	Requirement	Complete?
☐	Problem definition and mission need evidence is complete and ready to be graded.	
☐	Research and requirements evidence is complete and ready to be graded.	
☐	Concept development evidence is complete and ready to be graded.	
☐	Design selection and justification evidence is complete and ready to be graded.	
☐	CAD / technical design evidence is complete and ready to be graded.	
☐	Prototype / model quality evidence is complete and ready to be graded.	
☐	Testing and data analysis evidence is complete and ready to be graded.	
☐	Iteration and risk response evidence is complete and ready to be graded.	
☐	Final portfolio documentation evidence is complete and ready to be graded.	
☐	Final design review evidence is complete and ready to be graded.	
☐	All pages/files are named, organized, and submitted correctly.	
☐	Reflection or revision notes are complete.	

Teacher
Use the
evidence
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